

Soil Organic Carbon Model Intercomparison

Bayesian Calibration Framework for Finnish Forests

Lorenzo

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Current State: Multi-Model Framework

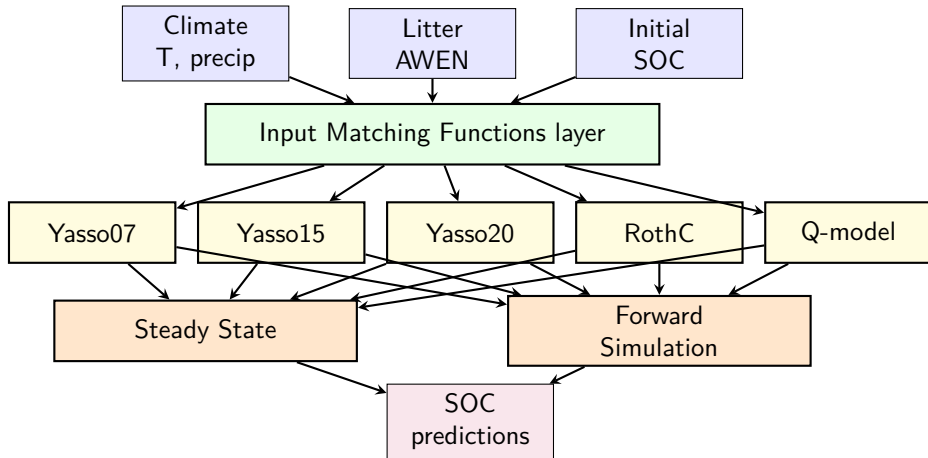
The models are so far **implemented in R** for faster prototyping. The code seems fast enough to run it in R via Bayesiantools, but multi-threading on Roihu needs still to be tested. Re-coding possible but prefer to avoid.

Five SOC Models Implemented & Debugged:

- **Yasso family:** Yasso07, Yasso15, Yasso20 (chemical fractionation)
- **RothC:** Decomposability-based pools
- **Q-model:** Continuous quality theory

Implementation:

- One layer of model-specific input rerouting for compatibility
- Analytical steady-state solutions for Yasso07, Yasso15, Yasso20 and RothC
- Optimized Q-model (Swedish hybrid init + Rcpp)



Key Optimizations: Q-Model

Challenge: Q-model steady state is a coarse approximation and causes differences in the first decade or two C balance. The initial cohort matrix needs to be initialized with numerical spin up, but this is computationally intensive.

Solution 1: Swedish Hybrid Initialization

- Combines analytical steady-state with numerical adjustment
- Avoids full numerical spinup from zero
- **Result:** 10–20x speedup while maintaining accuracy

Solution 2: Rcpp Compilation

- Performance-critical components compiled to C++
- Maintains R interface for integration
- Essential for feasibility of full-scale calibration

Example: Running the Multi-Model Wrapper

```
results_default <- run_soc_models_multipplot_calibration(  
  input_df = input_data,  
  site_df = site_data,  
  models = c("yasso07", "yasso15", "yasso20", "rothc", "q_model"),  
  params_list = NULL, # NULL means use defaults  
  spinup_years = 100  
)
```

Parameter values can be passed to the wrapper:

```
results_custom <- run_soc_models_multipplot_calibration(  
  input_df = input_data,  
  site_df = site_data,  
  models = "q_model", # Only run Q-model  
  params_list = list(q_model = custom_q_params), # Custom params  
  spinup_years = 100  
)
```

Bayesian Calibration Framework

Successful Pilot Runs:

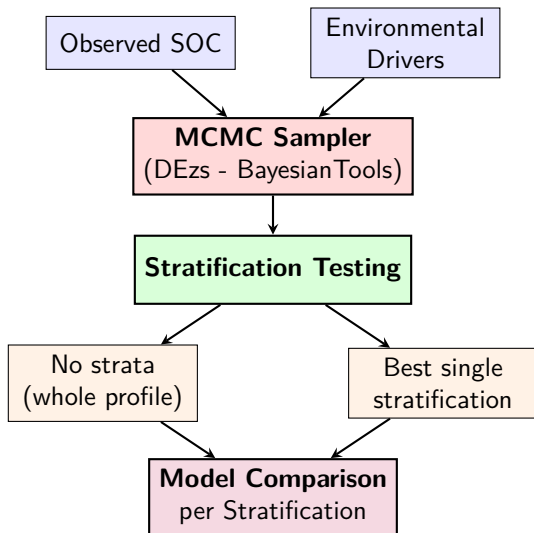
- BayesianTools package (DEzs sampler as default)
- Good convergence on small datasets
- Proper parameter passing validated
- Tested locally on Mac

Calibration Strategy:

- Models run in **independent batches**, not joint distribution
- Batches will be independent executions
- Per-plot calibration across full dataset

Scaling Path: - Local testing → **Directly to Roihu cluster** - Parallel processing across forest plots

Future Sampler Implementation



Strategy: Test models as-is with different stratification approaches

Planned Analysis Approach

Test each model (Yasso07, Yasso15, Yasso20, RothC, Q-model) with:

- **No stratification:** Whole soil profile integration
- **Single stratification:** Best horizon split to test

Models tested as-is

Models are kept as much as possible close to original formulations, and differences between them are therefore many and impossible to disentangle.

Goals:

- 1 Identify whether stratification improves model fit across models
- 2 Ensemble uncertainty predictions
- 3 Compare different modeling approaches (although not very rigorously)

Scaling Approach:

- Local testing (Mac M-series) for validation on 3-4 plots
- Then deployment to Roihu cluster
- Parallel processing across all plots in independent model batches for risk mitigation

Expected Challenges:

- **Runtime:** Full calibration across all plots $\times$ 5 models = at least **one day** of execution **per batch**
- This with 2000 steps chains. Convergence might be off.
- **Factorial design might be computationally unfeasible!!**

Forest time series (MONTHLY!):

- AWEN chemical fractions
- Distinct AWEN fractions for: Foliage, Branches, Fine Roots, Coarse Roots, Understorey
- Repeated SOC measurements across Finnish forests

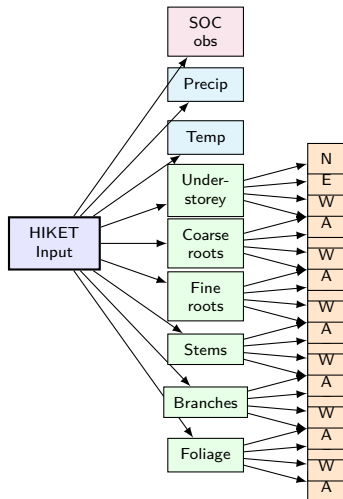
Forest time series (CONSTANT):

- Species composition: Pine, Spruce, Birch (for Q)
- Clay (for RothC)

Environmental Drivers (MONTHLY!):

- Monthly temperature and precipitation

Data requirements



Climate Scaling Implementation Across Models

Model	Time scale	Temperature Response	Moisture Response
Yasso07	Annual	Gaussian (mean, range)	Precipitation-based annual modifier
Yasso15	Annual, pool specific	Gaussian (mean, range)	Precipitation-based annual modifier
Yasso20	Monthly, pool specific	Gaussian (mean, range)	Precipitation-based annual modifier
RothC	Montly	Arrhenius	Moisture deficit model (rain, evap, SMC)
Q-model	Annual	Linear	Absent

For Q a quality dependent scaled Arrhenius is available, but not implemented yet.

Difficulties comparing the models

The models all rely on very different:

- Structures
- Response functions
- Input assumptions
- Calibration assumptions

Ideally we would test these **factorially** (as originally discussed), but I don't think HIKET scope allows for that. . .

Open questions

- Stratification levels?
- Is it fine to skip factorial model structure comparison?
- Discuss initialization and input unification approaches
- Validation? Crossvalidation or leaving a set out? I am more for the latter. . .

Status:

- The model wrapper is ready for forward runs

Next steps:

- 1 Forward runs without calibration, comparison with measurements
- 2 Base model comparison results for policy (mid-project deliverable)
- 3 Full-scale calibration against measured SOC run on cluster across all plots
- 4 Calibrated model comparison results for policy (final deliverable)

There might be hiccups on the way.